Graphpad Prism tutorial

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The University also acknowledges and is grateful to the Traditional Owners, Elders and Knowledge Holders of all Indigenous nations and clans who have been instrumental in our reconciliation journey.

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We pay respect to Elders past, present and future, and acknowledge the importance of Indigenous knowledge in the Academy. As a community of researchers, teachers, professional staff and students we are privileged to work and learn every day with Indigenous colleagues and partners.

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Let’s get started!

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What we'll cover today

- What kinds of things can GraphPad Prism do, with a focus on graphs
- How to import data into GraphPad
- How to make specific graphs and run appropriate associated analyses
- How to customise your figures
- How to export figures and tables
- It will be a mix of slides, live demonstrations, and working on examples together
- We will stop and run through some worked examples together
  - Don't worry if you don't get through the examples in time
  - We will need a bit of background to get started, but the goal is to *learn by doing*
- Ask questions as they come up* (if time is tight, save them until the end)
- Slides will be available after the seminar ☺
- We won't cover *everything* – this is all about getting started 🚀🏁▶️

What is GraphPad and why use it?

- GraphPad is a general-purpose statistics and graphing suite
- Pros:
  - Menu-based interface (no coding required)
  - Can be used to prepare publication-quality graphics and statistical analyses
  - Good online documentation, also thorough guide to the statistics supported
  - Analyses you run link back to the online Statistics Guide (e.g., explaining assumptions)
- Cons:
  - Price (may or may not be an issue)
  - Supports the commonest statistical analyses, but not everything
  - Type of plot/analysis affects how you input the data (more on this later)

Software considerations

- Cost: Free 30-day trial, or $300 per device per year* for the full installation
  - Go to the Staff Hub, search for Graphpad Prism, follow instructions on the first link
  - *It may be less than this, based on pro-rata licensing periods
  - Some features are <u>locked</u> 🙄, requiring upgrade to Pro/Enterprise plan
    - Locked features not needed for commonest use-cases
      - In-Table Formulae are possibly the most useful locked feature; can be done in Excel
- Available for Window, Mac and cloud
- Alternatives:
  - Minitab: Similar functionality, probably easier to use, similar price (*for UoM staff)
  - SPSS: Also menu-based, more statistics options, centrally funded for UoM staff
  - R: Analyses/plots set up with code (not menus/dialogue boxes), steeper learning curve, free, every statistical method under the sun…
    - Full disclosure: I use R for all my work.

A little about me and the SCC

- I've worked in data analysis, statistics, numerical modelling for >20 years
  - Mostly in environmental science, bioinformatics, biostatistics
  - Formerly an academic, now a consultant at the Statistical Consulting Centre
- We provide advice and support in data analysis, stats, visualisation, etc, etc ☺
  - For eligible* Masters/PhD students, 10 free hours, then paid (discounted) support
  - For academics, a fee-for-service model at cost-recovery rates
  - We support clients outside of the University on a commercial basis
  - We also run short-courses, and seminars
    - Introductory R course in July (in person)
    - Statistics for Research Workers in Nov/Dec (in person)
    - A seminar on producing quality graphics in R and Minitab (on Zoom)
    - Monthly seminars (second Friday of each month at 12:30, on Zoom)
  - Find out more at https://scc.ms.unimelb.edu.au/

Overview of the Prism interface

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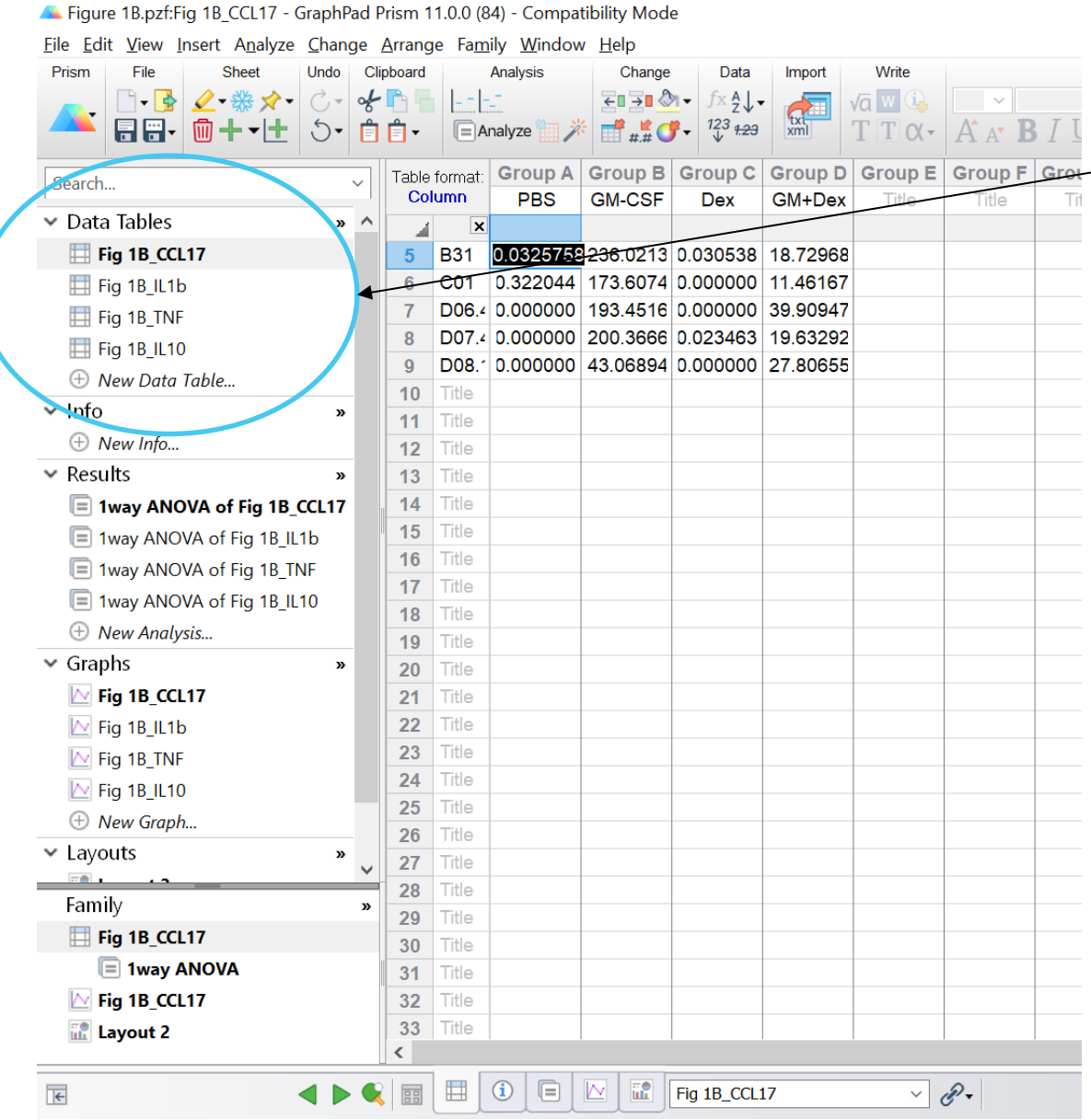
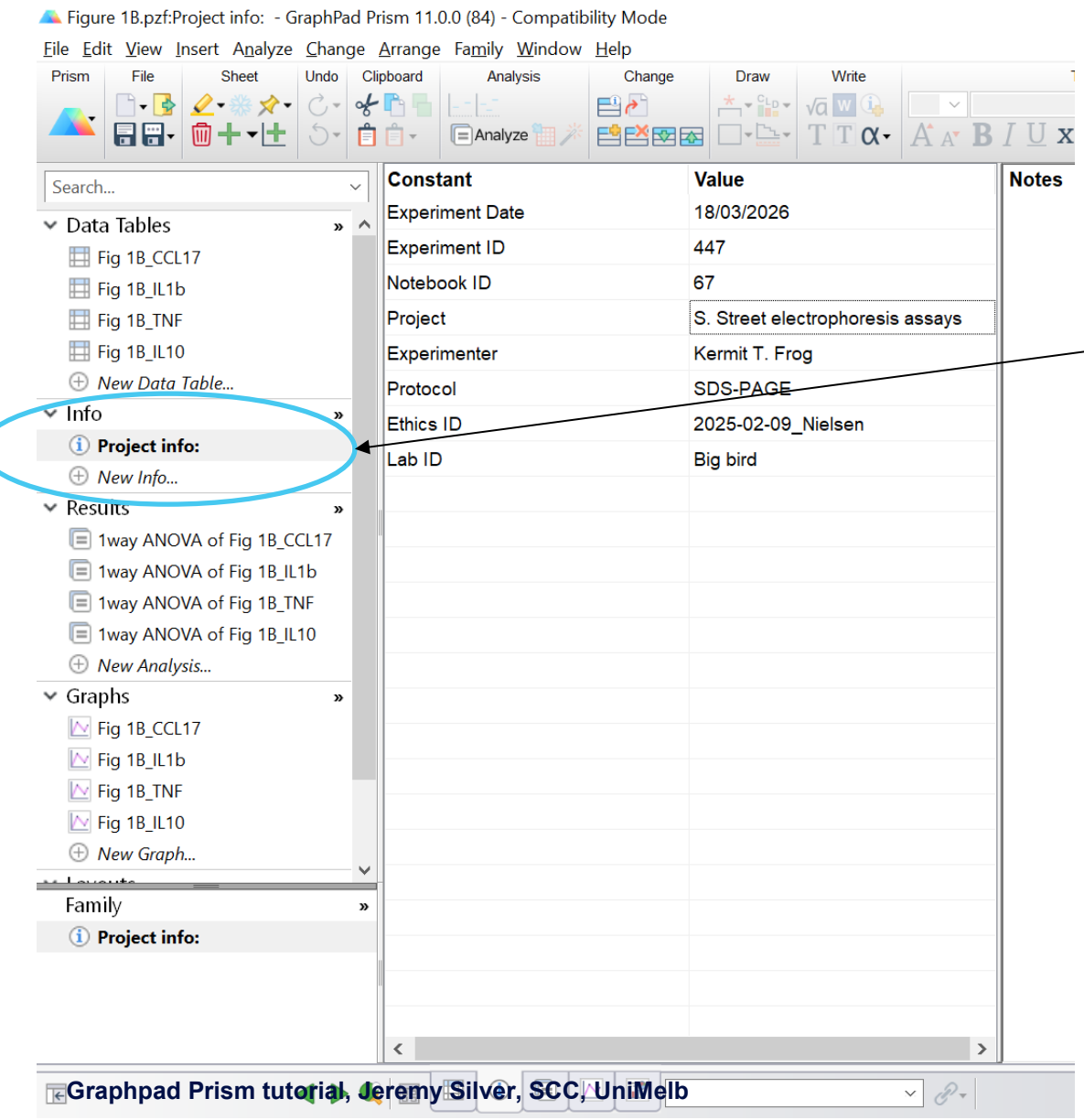
Prism interface: Data tables



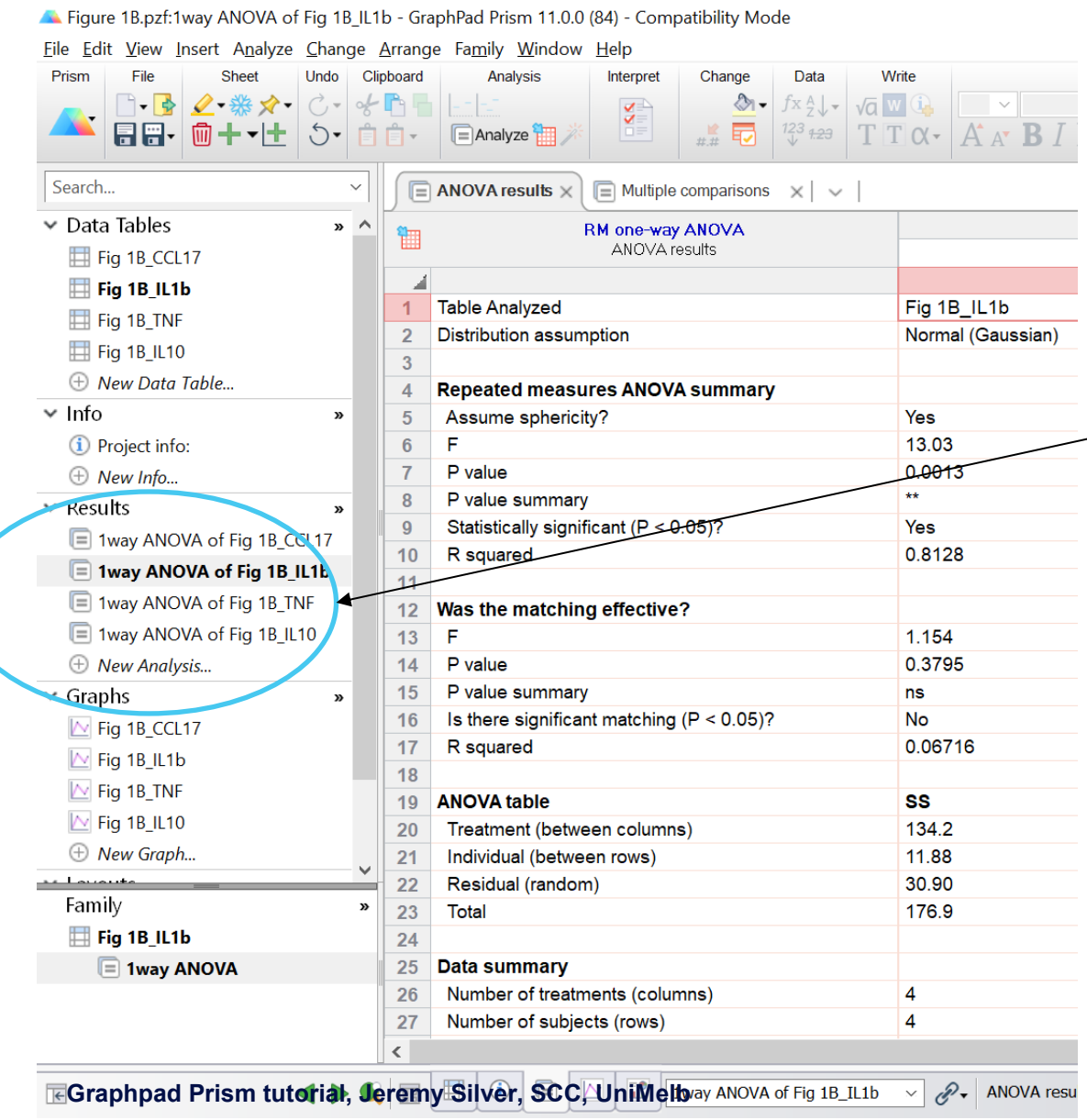
These are like tabs in an Excel spreadsheet

Prism interface: Info



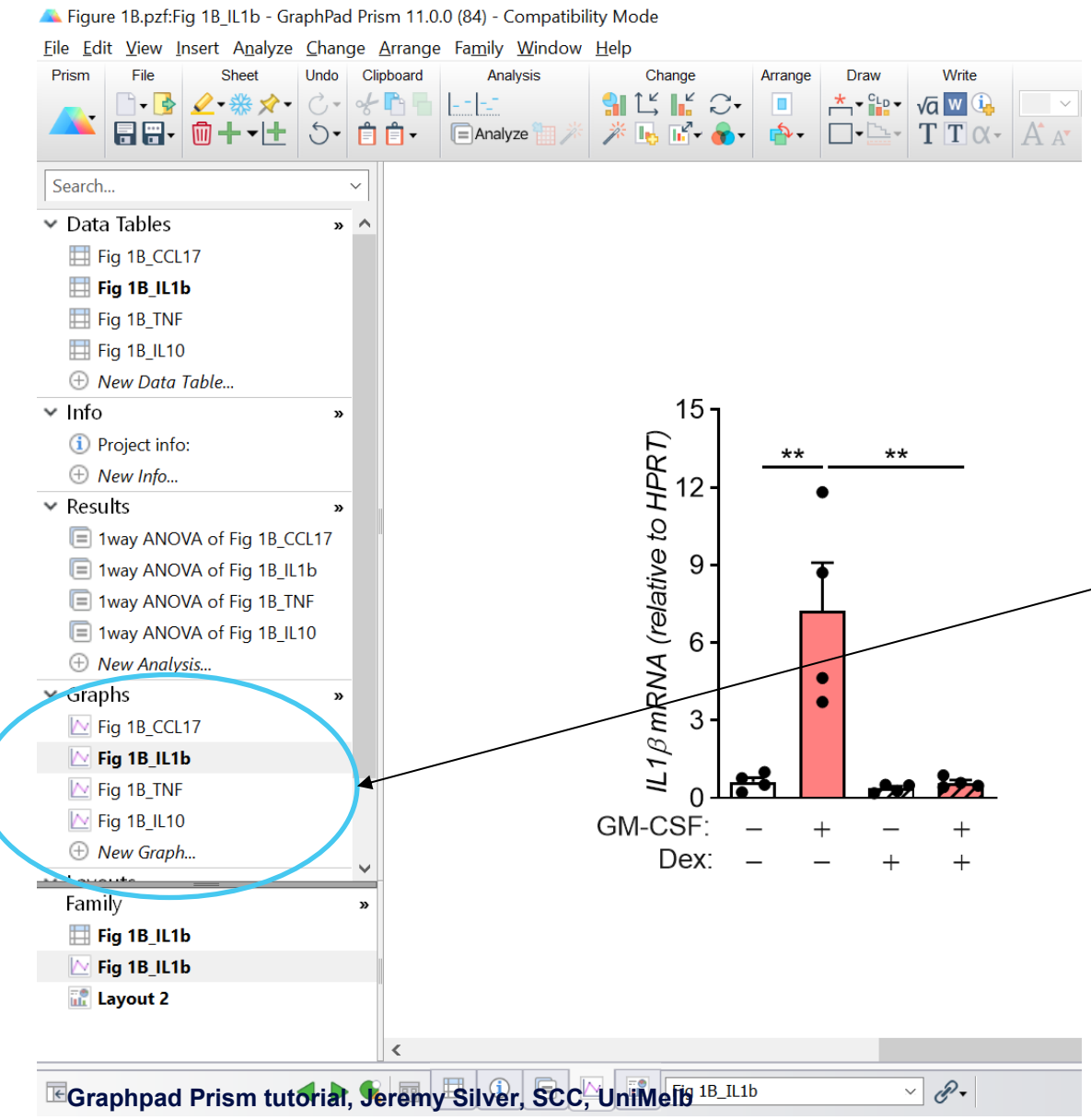
You can record info about the data source/experiment here

Prism interface: Results



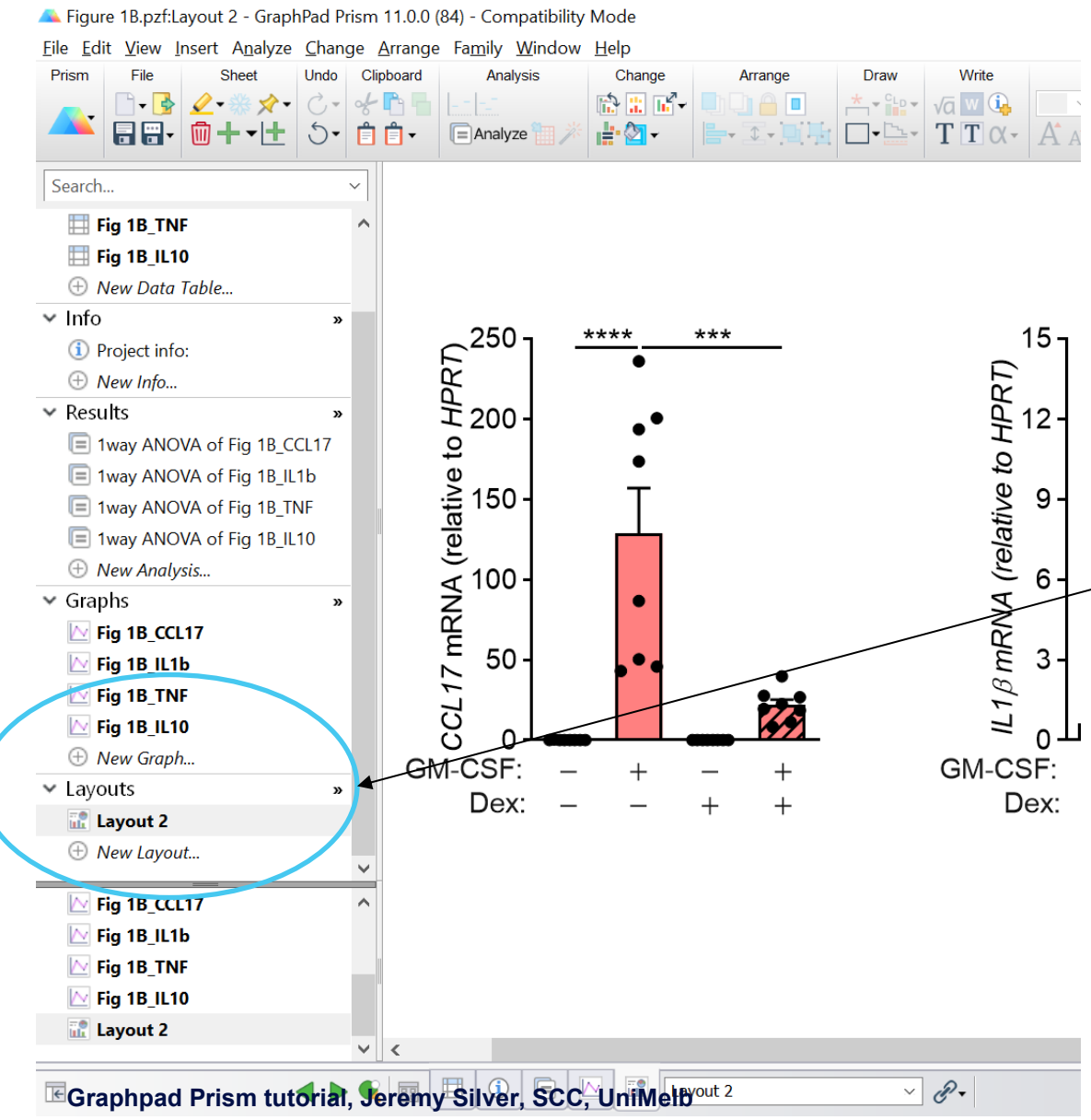
When you run statistical analyses, they appear here

Prism interface: Graphs



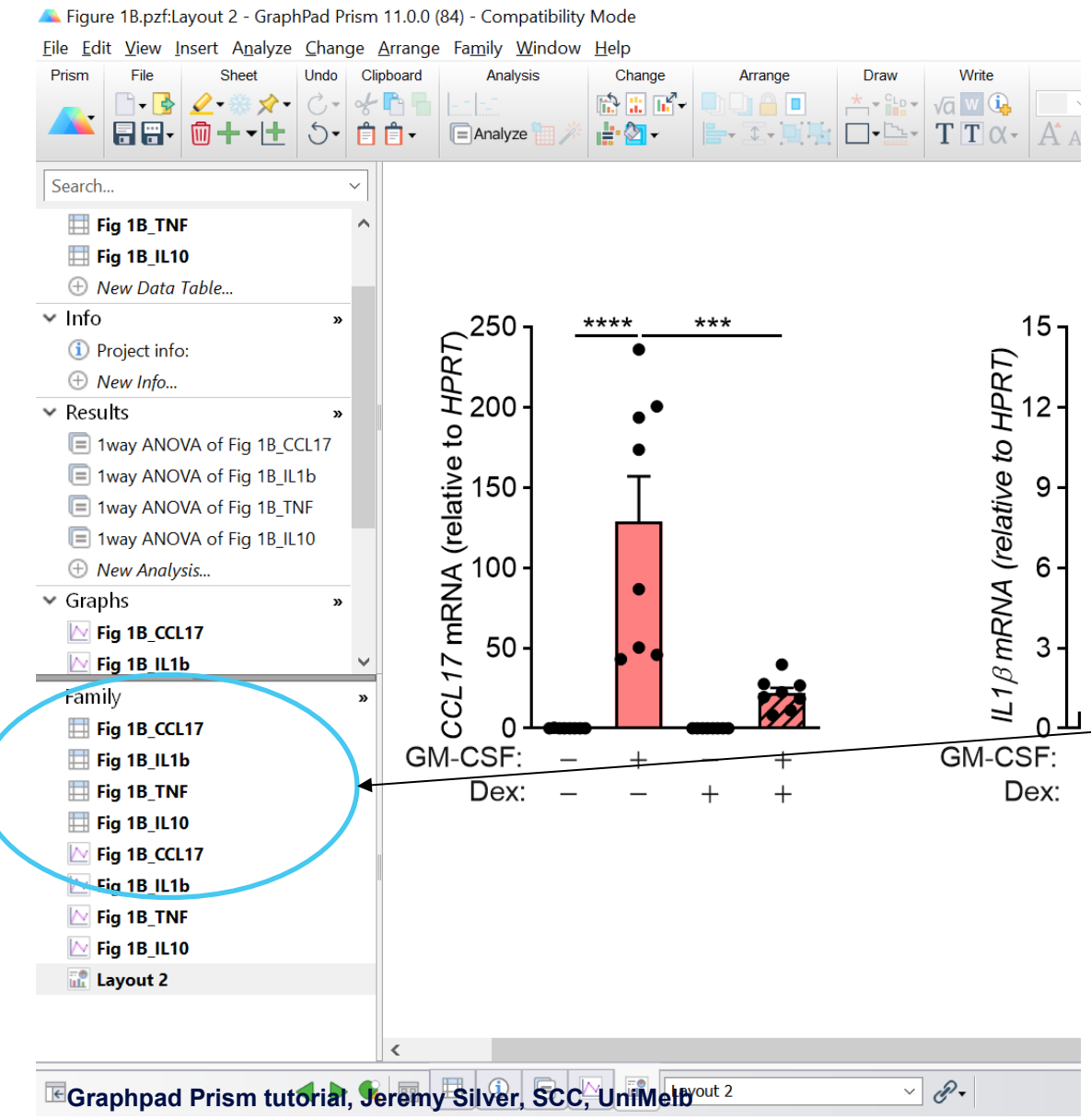
Graphs that you make appear here. When you add a Data table, a graph will be made.

Prism interface: Layouts



The “Layouts” section allows you to place your figures together in a multi-panel layout.

Prism interface: Family



The “Family” section shows you how everything that is connected to what you are currently viewing: data tables, plots, analyses, layouts.

If you have lots going on, it can be a handy way of navigating between related components.

The thinking behind Prism

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You need to have a plan 📝🎯

- In Prism, certain things are interlinked:
  - Data organisation (what kind of “Data table” you have)
  - What analyses you can do
  - What plots you can make
- Logical, but frustrating…😤
- Before you enter your data, you *must* choose a Data Table type
  - So, you need to know a bit about what the Data Table types are, and what they permit.

Prism's 8 Data Table types 😕

- XY tables
  - One *x* variable (e.g., time, concentration), one or more values of a single response variable
  - There can (optionally) be a grouping variable
- Column tables
  - No natural *x* variable, a single response variable, there is *one* grouping variable
- Grouped tables
  - No natural *x* variable, single response variable, there are *two (or three)* grouping variables
  - Effects are *crossed* (e.g., drug treatment by age-group)
- Nested tables
  - No natural *x* variable, single response variable, within a grouping variable
  - Similar to Grouped Tables, but subgroups are *nested* (not *crossed*)
  - e.g. a treatment is applied to all mice in a cage, mouse cages are nested within treatments

Prism's 8 Data Table types 😕

- Contingency tables
  - Counts of instances, broken down by different categorical factors
- Parts-of-whole tables
  - It's what it says on the tin… Breaking totals into their fractional components
- Multiple variables tables
  - Each column is a different variable, variables can be continuous or categorical
  - Most other statistics programs arrange data this way
- Survival tables
  - A time-to-event-or-censoring variable, and an event-or-censoring indicator
  - Optional: grouping by a categorical factor

Let’s dive in!

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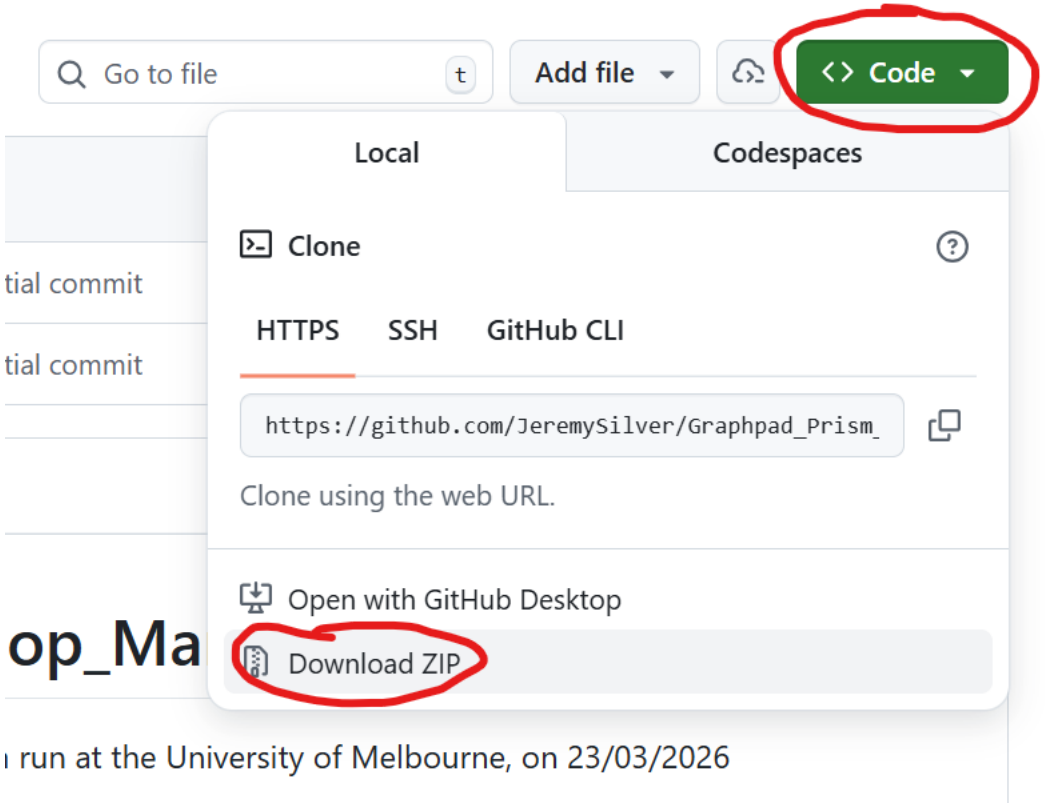
What we will try to do

- Load in data, produce descriptive statistics, analysis and plots for:
  - A two-group comparison (e.g., *t*-test, Mann-Whitney test)
  - A multiple group comparison (e.g., one-way ANOVA)
  - Accounting for multiple groups simultaneously (e.g., two-way ANOVA)
  - A simple linear regression
  - Correlations between variables (e.g., Pearson, Spearman correlations)

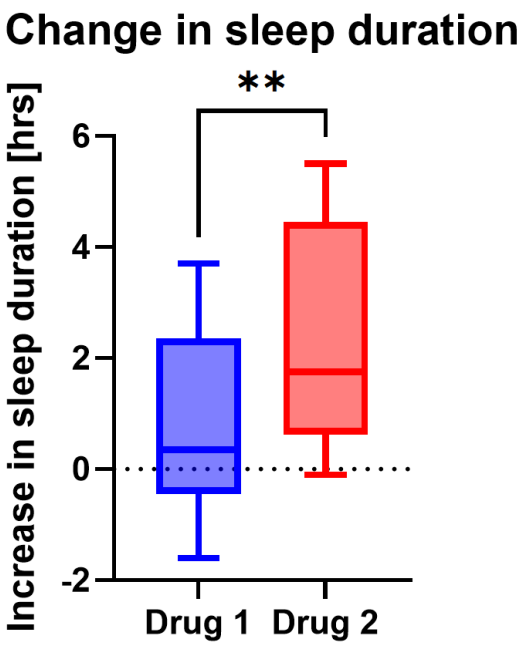
Let’s get some data to play around with

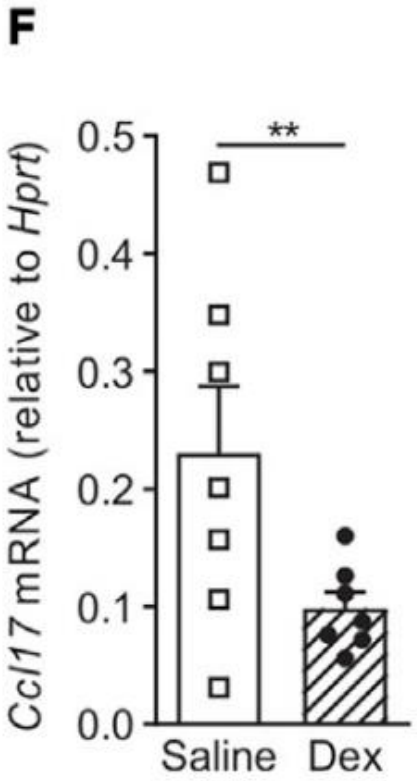
- https://github.com/JeremySilver/Graphpad_Prism_workshop_March_2026
- Click on the green “<> Code” button, then “Download ZIP” from the menu



What we will try to do

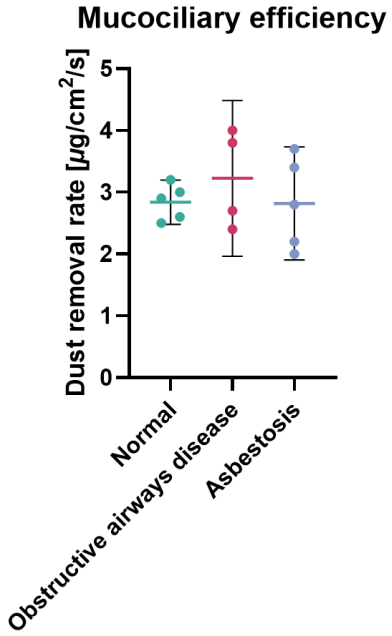
- Load in data, produce descriptive statistics, analysis and plots for:
  - A two-group comparison of means (e.g., *t*-test, Mann-Whitney test)
    - Start with a “Column table”
    - Live demo example: “sleep study”
    - Self-study example “Fig 4F”

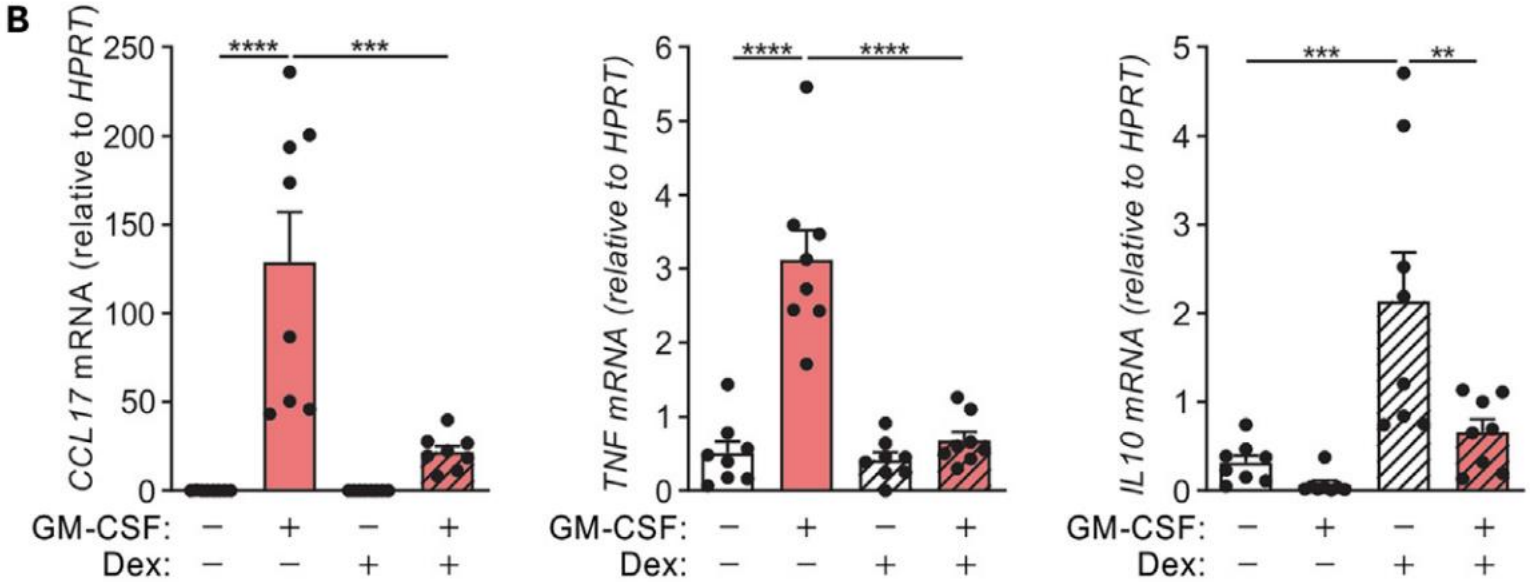




What we will try to do

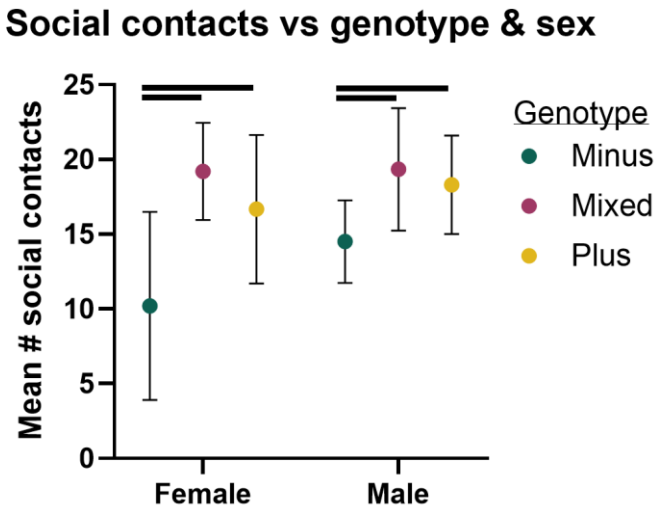
- Load in data, produce descriptive statistics, analysis and plots for:
  - A multiple group comparison of means (e.g., one-way ANOVA)
    - Start with a “Column table”
    - Live demo example: “Mucociliary efficiency” (dust removal)
      - three_group_comparison_data.csv
    - Self-study example “Fig 1B”

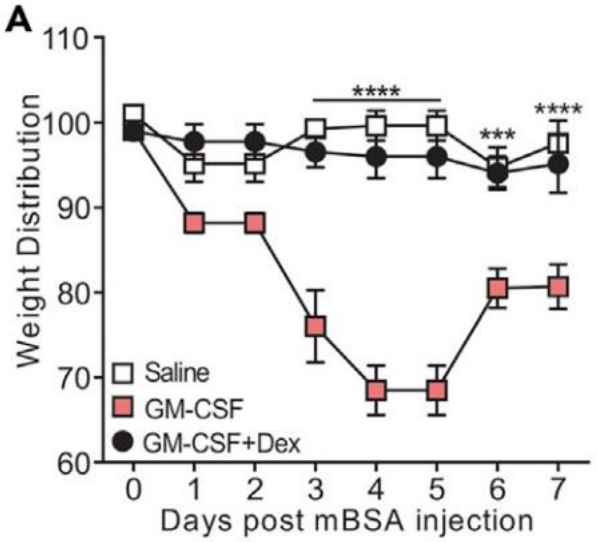




What we will try to do

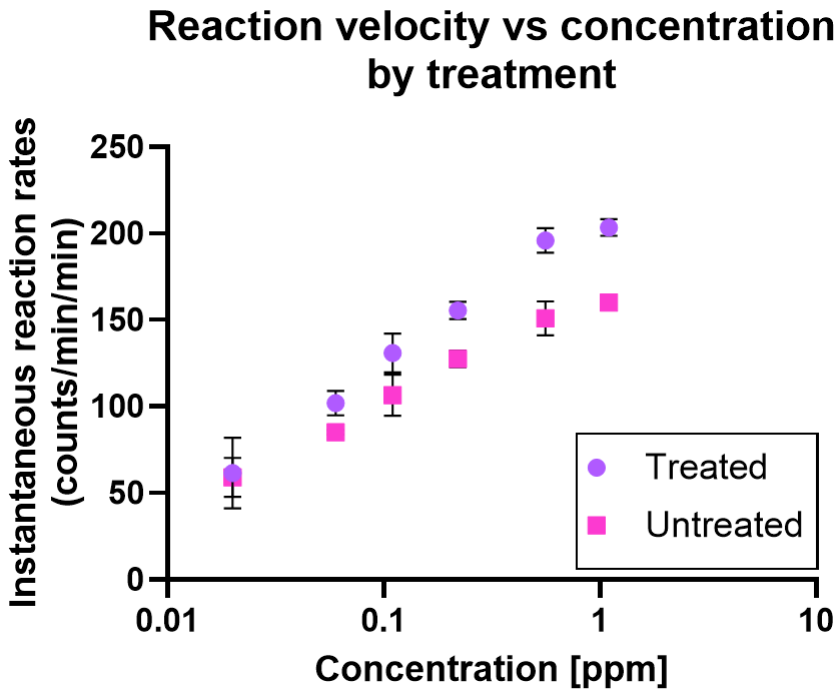
- Load in data, produce descriptive statistics, analysis and plots for:
  - Accounting for multiple groups simultaneously (e.g., two-way ANOVA)
    - Start with a “Grouped table”
    - Live demo example: “MouseBrain.csv” (or MouseBrain_reformatted_for_Prism.csv)
    - Self-study example “Fig 4A”

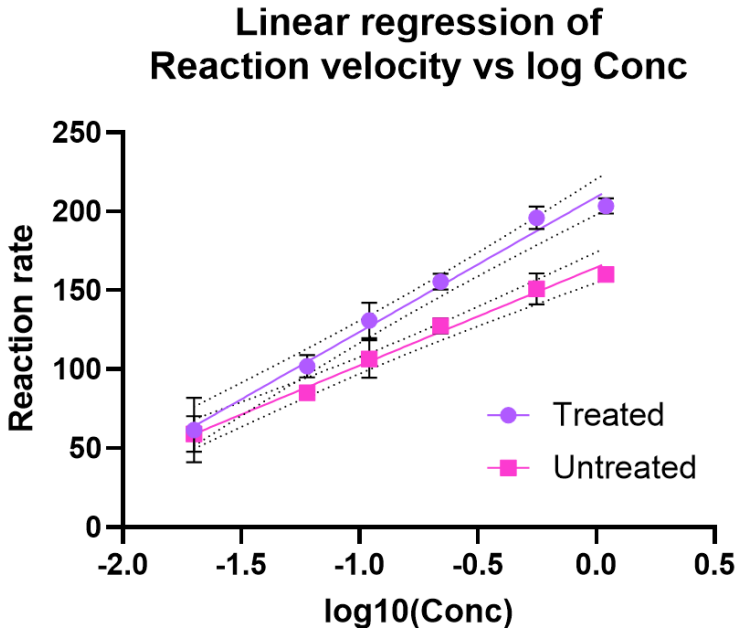


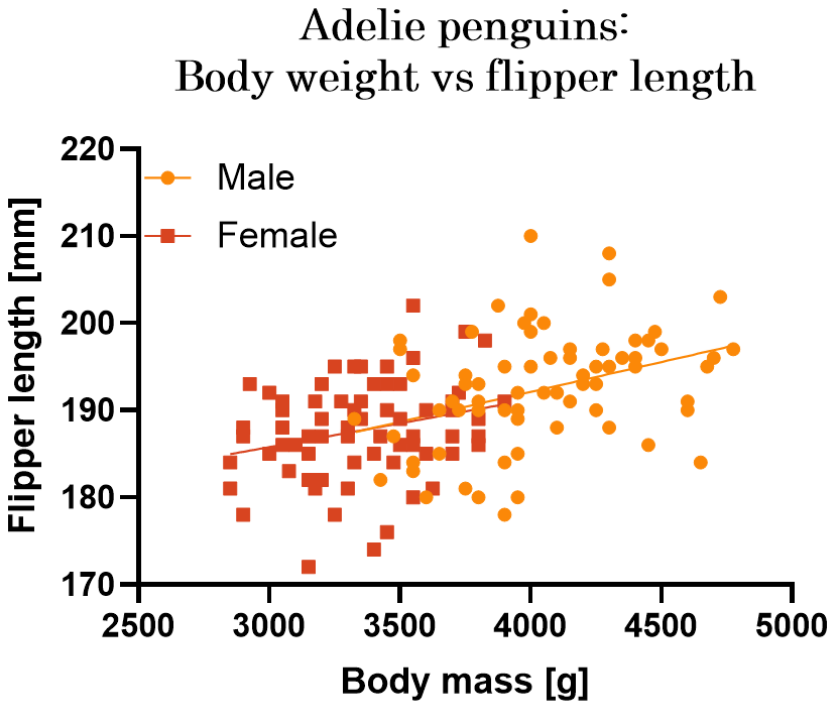


What we will try to do

- Load in data, produce descriptive statistics, analysis and plots for:
  - A simple linear regression
    - Start with a “XY table”
    - Live demo example: “Puromycin” data
    - Self-study example “penguins” data (just Adelie penguins; body weight vs flipper length)

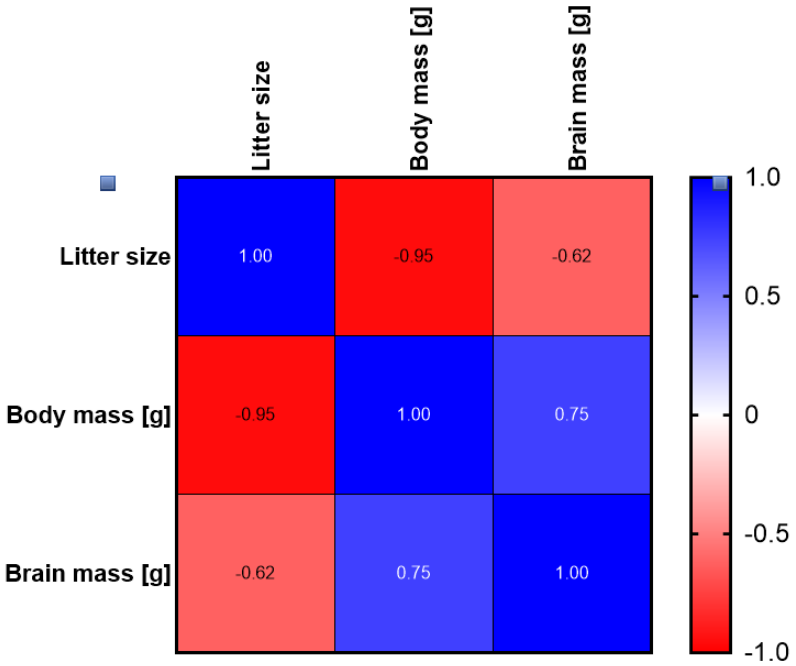


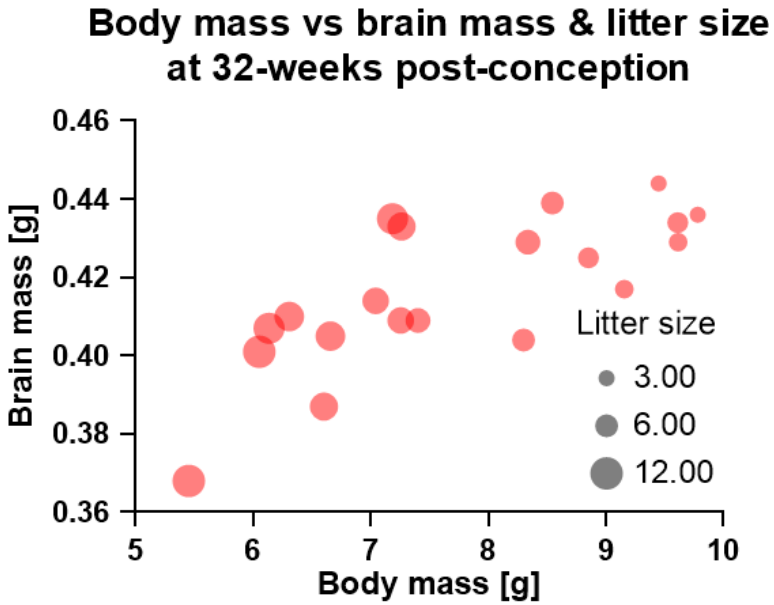


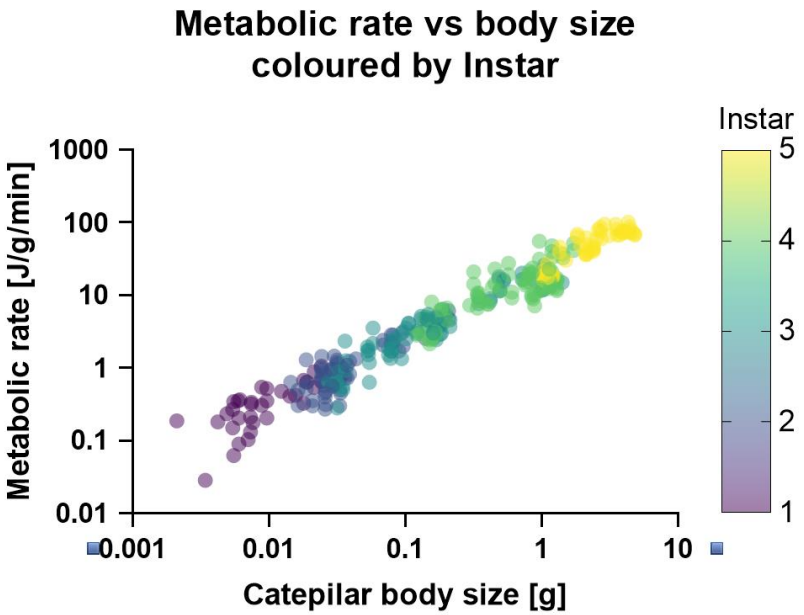


What we will try to do

- Load in data, produce descriptive statistics, analysis and plots for:
  - Correlations between variables (e.g., Pearson, Spearman correlations)
    - Start with a “Multiple Variable table”
    - Live demo example: “Mouse litters” (litters.csv)
    - Self-study example “Metabolic rate”







Wrap up

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Key things to remember

- Need to decide what plots/analyses you do *before* getting started
  - This affects the choice of data table type, which limits options
  - But you can analyse the data in different types of data tables if needed
- Getting data in the right format for entering can be tricky
  - Reformatting in Excel may be simpler (try “Paste special” → “Transpose”)
- Descriptive statistics are really helpful – generate these routinely
- GraphPad’s online Statistics Guide and Prism Guide are really helpful
  - Also, the GraphPad “Resources” page has a lot of useful video tutorials
- Once you have run an analysis, try the “Interpret” button:
  - Links to the page in the Statistics Guide, clarifies assumptions

- With graphs, you can edit/move most plot components by clicking on them

Finally, don’t hesitate to reach out to us 😊🤝

- We provide advice and support in data analysis, stats, visualisation, etc, etc ☺
  - For eligible* Masters/PhD students, 10 free hours, then paid (discounted) support
  - For academics, a fee-for-service model at cost-recovery rates
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